

A δ^2 – L^3 Sensitivity Law for Finite-Window Weil Positivity, with a Function-Field Control of the Connes–van Suijlekom Pipeline

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Abstract

The Connes–van Suijlekom and Connes–Consani–Moscovici programs reduce the Riemann hypothesis to spectral properties of a quadratic form restricted to test functions supported in a finite window. We ask how sensitive that form is to a zero moved off the critical line, and we calibrate the answer in a setting where the truth is known: curves over finite fields, where the window form is exactly the Frobenius Gram matrix of Hallouin–Perret and the Riemann hypothesis is Weil’s theorem. Our main result is a sensitivity law. In the function field we prove that moving a reciprocal real pair $\{\rho, \rho^{-1}\}$ off the critical circle gives $\lambda_{\min}(T_R) = -\binom{R+2}{3}a^2 + O(a^4)$ with $a = \log \rho$, the coefficient being twice the centred second moment of the window. On the zeta side, in the even sector, we prove an bound $\lambda_{\min}(Q_\delta) \geq \lambda^*(L) - c(L)\delta^2 - O(\delta^4)$ with $c(L) = 8L^3(\frac{1}{3} + \frac{1}{\sqrt{5}})$, stated *relative* to a lower bound $\lambda^*(L)$ for the unperturbed form and unconditional where that positivity is certified ($L \leq 0.8$), and a sharper conditional bound with the *exact* constant $4K(L, \gamma)$, where $K(L, \gamma) = \int_{-L}^L u^2 \sin^2(\gamma u) du$ is attained by $f \propto u \sin(\gamma u)$; numerically the re-optimised minimiser reaches up to 96% of it. Under an explicit dictionary the two laws are the same expression, (window length)³/6. We also run the full Connes–van Suijlekom minimiser pipeline on deliberately planted off-circle spectra, which separates the steps that carry arithmetic content from those that do not. **Nothing here proves or advances the Riemann hypothesis for ζ .**

Non-claims

Stated first, not in an appendix.

1. **No proof or advance of RH for ζ .** Part I runs where the Riemann hypothesis is a theorem (Weil, curves over finite fields); Part II uses fabricated spectra.
2. **No new spectral architecture.** The window form, its Toeplitz structure and its Carathéodory–Fejér theory are due to Hallouin–Perret [8] and Connes–van Suijlekom [5]; the finite-window Weil form itself goes back to Yoshida [10] and Bombieri [1].
3. **Planted spectra are fabricated diagnostics**, not L -functions of anything.
4. **Numerical detections are tier T2 unless certified**, and *non-detections certify nothing*: a finite basis that finds no negative direction is not evidence of positivity. This asymmetry is the subject of §6.

5. $\lambda^*(0.8) \leq 2.2702 \times 10^{-17}$ **is a variational upper bound** from a finite basis, not a certified enclosure. We prove no rigorous lower bound *of our own*; Zhu [11] certifies one, and Theorem 7 uses it to convert its relative estimate into an unconditional lower bound for $L \leq 0.8$. Theorem 8 does *not* use it; it rests on its own hypothesis.
6. **§5 reproduces published computations** (Groskin [6]) at smaller scale.
7. **Block F is claimed only for the function field.** On ζ windows the indefiniteness of A and M depends on where $-\log \pi$ is placed; see §7 and Retraction 15.
8. **Not attempted:** Davenport–Heilbronn; any Lean formalisation.

Epistemic tiers

Tier	Meaning	Discharged by
T1	exact or proved	a proof, or exact arithmetic over $\mathbb{Z}$ or $\mathbb{Q}$
T2	numerical	stated working precision + a convergence or noise-floor check
T3	interpretation	argument from T1/T2 evidence, labelled as such

Every numbered statement below carries a tier, the script that produced it and the data file that records it. All numerics use `mpmath` at the stated precision; `float64` is used only as a deliberate contrast (§10).

1 The pipeline and the question

Let $X/\mathbb{F}_q$ be a smooth projective absolutely irreducible curve of genus g , and let $N_n = \#X(\mathbb{F}_{q^n})$. Hallouin and Perret [8, Prop. 5] show that the normalised classes γ^k of the iterated Frobenius graphs have Gram matrix

$$\text{Gram}(\gamma^0, \dots, \gamma^R) = (t(|i - j|))_{0 \leq i, j \leq R}, \quad t(0) = 2g, \quad t(n) = \frac{(q^n + 1) - N_n}{q^{n/2}}, \quad (1)$$

a symmetric Toeplitz matrix, which we write T_R . We adopt (1) as the definition of the window form; it is *their* matrix, entry for entry.

The Connes–van Suijlekom pipeline [5] passes through four steps: (i) reality of the zeros of the ground-state polynomial; (ii) the sign of $\lambda_{\min}$; (iii) simplicity and evenness of the ground state; (iv) convergence of the normalised minimiser. The question of this note is *quantitative*: how large a negative eigenvalue does a zero moved a distance δ off the critical line actually produce, and how large a window is needed to see it?

What is claimed as new. Only two things.

(N1) The δ^2 – L^3 sensitivity law: Theorem 6 (function field), Theorems 7 and 8 (ζ), the measured near-saturation, and the shared (window length)³/6 form.

(N2) The planted-spectrum control of the CvS minimiser pipeline in the function field (§3).

Everything else is foundation or confirmation; §8 says exactly whose.

2 Foundation (prior art, stated as such)

Proposition 1 (Hallouin–Perret [8, Prop. 5, §1.2]). *[T1, cited] The Gram matrix of the normalised iterated-Frobenius classes is the Toeplitz matrix (1).*

Theorem 2 (Hallouin–Perret [8, Thm. 6, §1.3]). *[T1, cited] Let d be the rank of the Frobenius space F . Then d is the minimal integer such that $\text{Gram}(\gamma^0, \dots, \gamma^d)$ is singular, a generator of that kernel exists over $\mathbb{Q}$, and the resulting Frobenius eigenvalues are algebraic integers of modulus $\sqrt{q}$.*

Theorem 3 (Hallouin–Perret [8, Thm. 36(ii), App. A.2]). *[T1, cited] If a real symmetric positive semi-definite Toeplitz matrix of size $n + 1$ has rank n and kernel vector $(a_0, \dots, a_n)$, then $P = \sum a_j X^j$ has n distinct roots, all of modulus 1.*

Theorem 3 is step (i) of the pipeline; it is the Carathéodory–Fejér corollary that CvS [5, Cor. 1.1] prove again C^* -algebraically. Their Lemma 33 (rank = largest non-zero *leading* minor) and Lemma 34 (the $(\gamma_i \pm \gamma_{n-i})/\sqrt{2}$ parity split) are the computational tools we use throughout. CvS themselves point to [8] for the function-field connection and write that “the key difficulty in this context, then, becomes the verification that zero is indeed the (simple) minimal eigenvalue of T ” [5, Introduction].

3 The control experiment

3.1 Genuine spectra

[T1 for the arithmetic, T2 for the spectra; `src/stage1.py`, `data/stage1_spectra.csv`] Four curves ($y^2 = x^3 + x/\mathbb{F}_5$, $y^2 = x^5 + x^2 + 1/\mathbb{F}_3$, $y^2 = x^7 + x + 1/\mathbb{F}_5$, $y^2 = x^7 + x^3 + x + 1/\mathbb{F}_3$; $g = 1, 2, 3$) with brute-force point counts, L -polynomials by Newton’s identities plus the functional equation, and exact integer power sums. All 40 rows have $\lambda_{\min} \geq 0$; $\lambda_{\min}(R)$ decreases monotonically (Cauchy interlacing) and vanishes exactly at $R = d$, where the kernel polynomial’s roots are the $e^{i\theta_j}$ — Theorem 2 in action.

3.2 Planted spectra

[T1, exact over $\mathbb{Q}$; `src/stage2.py`, `src/exact_inertia.py`] We replace the Frobenius spectrum by a functional-equation-closed multiset with $|\beta| \neq 1$. Choosing rational ρ and $\cos \varphi$ makes every $t(n)$ rational, so the inertia is certified *exactly* by the Sylvester–Jacobi leading-minor rule.

configuration	planted	first R with $\lambda_{\min} < 0$	step (i) holds?	$\lambda_{\min}$ simple?
on-line control	$ \beta = 1$	never (PSD, proved)	always	fails for every $R > d$
real pair	$\rho = 1.3$	$R = 1$	always	never fails
quartet	$\rho = 1.3, \varphi = 0.7$	$R = 2$	always	never fails
two quartets	$\rho = 1.3$	$R = 4$	always	never fails
mixed	quartet + pair	$R = 4$	always	never fails

Observation 4. [T1 reasoning, T2 evidence] Step (i) holds at every applicable row of every planted configuration, including rows where $\lambda_{\min} = -642.8$: measured $\max||z| - 1| \leq 5.1 \times 10^{-48}$. Steps (i) and (iii) are therefore free of arithmetic content. This is not a discovery — CvS state Theorem 3 for an arbitrary real even distribution, so no arithmetic can enter step (i) — but the planted control makes it concrete, with numbers attached.

3.3 Degeneracy beyond the window degree

[T1 corollary of Thm. 3(i); `src/function_level.py`] Since the shift has P as both minimal and characteristic polynomial, $\ker T_R$ is the ideal generated by P truncated at degree R :

$$\ker T_R = \{P \cdot q : \deg q \leq R - d\}, \quad \dim \ker T_R = \max(0, R + 1 - d), \quad (2)$$

matched in every computed row. Hence $\lambda_{\min} = 0$ is simple exactly at $R = d$ and never after, so the CvS simple-ground-state hypothesis fails for every $R > d$; and since P is self-reciprocal, the kernel splits into even and odd parts of dimensions $\lceil \frac{m+1}{2} \rceil, \lfloor \frac{m+1}{2} \rfloor$ with $m = R - d$, both non-zero for $m \geq 1$. CvS state this in words in their Remark 2.3; we record the dimensions.

3.4 Step (iv) at the function level

[T1; src/function_level.py]

Proposition 5. *If $\lambda_{\min}$ is simple for infinitely many R and the normalised ground-state polynomials converge locally uniformly to $F \not\equiv 0$, then every zero of F lies on $|z| = 1$ by Theorem 3 and Hurwitz’s theorem. Hence they cannot converge to a target with a zero off the circle.*

The free witness is exactly the obstruction. Measured: zero *angles* converge to $\sim 10^{-3}$ even for off-line spectra, while the coefficient distance to the target is exactly $\sqrt{2}$ (orthogonal) and the normalised $|\hat{c}_R(\beta)|$ *rises* to 0.58–0.71 out to $R = 36$ instead of falling. This makes concrete the Hurwitz step that is CvS’s own Step 5.

4 The δ^2 – L^3 law

4.1 Function field

Theorem 6. [T1; proof in notes/proofs.md, checked by src/proofs_check.py] *Let the normalised spectrum be a single off-circle real pair $\{\rho, \rho^{-1}\}$ and put $a = \log \rho$. With $u_i = \rho^i$, $w_i = \rho^{-i}$ ($0 \leq i \leq R$),*

$$T_R = uw^\top + wu^\top, \quad \lambda_{\min}(T_R) = (R+1) - \|u\| \|w\|,$$

all other eigenvalues being 0. Consequently $\lambda_{\min} < 0$ for every $R \geq 1$ as soon as $\rho \neq 1$, by strict Cauchy–Schwarz; $\lambda_{\min}$ is an even function of a ; and

$$\lambda_{\min}(T_R) = -2a^2 \sum_{i=0}^R \left(i - \frac{R}{2}\right)^2 + O(a^4) = -\binom{R+2}{3} a^2 + O(a^4).$$

For $R = 1$ this is exact: $\lambda_{\min} = -4 \sinh^2(a/2)$.

The coefficient is twice the *centred* second moment of the window; the centring falls out of the optimisation, not from a convention. Checks: the factorisation to 6.2×10^{-61} , the eigenvalues to 1.0×10^{-59} , the moment identity exactly for $R = 1, \dots, 199$, and $-\lambda_{\min}/a^2 = \binom{R+2}{3}$ to twelve printed digits for $R = 1, 2, 4, 8, 16, 32$.

4.2 Zeta side

Throughout this subsection f is real, **even**, supported in $[-L, L]$ with $\|f\|_2 = 1$, and $F = \hat{f}$; all statements here are for the even sector.

The baseline, stated explicitly. An off-line zero forces a quartet — four zeros — whereas a *simple* on-line zero at γ contributes a pair. The zeros of ζ are expected to be simple, and every computed one is, so a count-preserving comparison must be made against a *hypothetical double* on-line zero. Writing Q_{true} for the actual Weil form, the baseline below is

$$Q_0 = Q_{\text{true}} + 2F(\gamma)^2. \tag{3}$$

Two consequences, both used. First $Q_0 \geq Q_{\text{true}}$, so Theorem 7 is unaffected: a lower bound certified for Q_{true} is also one for Q_0 . Second, Theorem 8’s hypothesis $\tilde{Q} := Q_0 - 4F(\gamma)^2 \geq 0$

reads $Q_{\text{true}} - 2F(\gamma)^2 \geq 0$, i.e. exactly “every zero *other* than the one being moved lies on the critical line”. Of the three conventions this is the only one with $Q_\delta \rightarrow Q_0$ as $\delta \rightarrow 0$, and (3) is what that continuity costs: it is bought by shifting the baseline, not by comparing against ζ itself.

Moving the double zero to $\frac{1}{2} + \delta \pm i\gamma$ changes the form by

$$Q_\delta(f) - Q_0(f) = -4\delta^2(F'(\gamma)^2 + F(\gamma)F''(\gamma)) + R_4. \quad (4)$$

The two baselines, the budget, and what λ^* denotes. Call f *admissible* if it is real, even, $\text{supp } f \subset [-L, L]$ and $\|f\|_2 = 1$. Write Q_ζ for the *actual* ζ Weil form on the window, so that $Q_0 = Q_\zeta + 2F(\gamma)^2$ by (3), and set

$$b_0(L, \gamma) = \inf_{f \text{ admissible}} Q_0(f), \quad \lambda^*(L) \leq \inf_{f \text{ admissible}} Q_\zeta(f),$$

so that b_0 is the infimum of the *augmented counterfactual* baseline and $\lambda^*(L)$ denotes any *lower bound* for the *actual* unperturbed form. These are two different objects and are nowhere identified. Both are infima of a quadratic form on the stated domain; nothing here assumes the Weil form is a bounded operator on L^2 . Put

$$B_L(\delta) = c(L)\delta^2 + \frac{16}{3}L^5\delta^4e^{2L\delta}, \quad c(L) = 8L^3\left(\frac{1}{3} + \frac{1}{\sqrt{5}}\right) \approx 6.244L^3,$$

the perturbation budget: $4|F'(\gamma)^2 + F(\gamma)F''(\gamma)| \leq c(L)$ by Cauchy–Schwarz and $|R_4| \leq \frac{16}{3}L^5\delta^4e^{2L\delta}$, both proved in §B.2–B.3 of `notes/proofs.md`.

A lower bound for Q_ζ on a larger test space is *a fortiori* a lower bound on the even sector, because an infimum over a larger space cannot exceed one over a subspace; so a certificate proved on the full admissible space may be quoted here unchanged. Zhu’s certificate is used only with its documented normalisation $Q(f)/\|f\|^2$ — the one used throughout — and only on its documented range.

Theorem 7 (relative perturbation estimate; even sector). *[T1] For every admissible f and every $\delta \geq 0$,*

$$Q_\delta(f) \geq Q_0(f) - B_L(\delta),$$

and hence, taking infima over the normalised admissible sector,

$$\inf Q_\delta \geq b_0(L, \gamma) - B_L(\delta) \quad \text{and} \quad \lambda_{\min}(Q_\delta) \geq \lambda^*(L) - B_L(\delta),$$

the second because $Q_0 \geq Q_\zeta$, so any lower bound $\lambda^(L)$ for the actual unperturbed form is also one for Q_0 .*

The estimate holds *relative* to the unperturbed lower bound $\lambda^*(L)$. Replacing this baseline by zero requires independently established nonnegativity. We use the cited positivity certificate for $L \leq 0.8$: Zhu [11] certifies $\lambda^*(0.8) \geq 8.9 \times 10^{-18}$ and λ^* is non-increasing in L , so

$$\lambda_{\min}(Q_\delta) \geq 8.9 \times 10^{-18} - B_L(\delta) \quad (L \leq 0.8).$$

No such replacement is asserted here for larger windows.

Validity versus usefulness of the budget. The remainder bound inside $B_L(\delta)$ is a convergent series tail, not a truncated asymptotic expansion, so $B_L(\delta)$ is a valid budget at every δ on its proved domain. What degrades as $L\delta$ grows is its *informativeness*: the δ^4 term overtakes $c(L)\delta^2$, after which $B_L(\delta)$ no longer reflects the quadratic law. Validity and usefulness are separate questions and only the second fails there.

Theorem 8 (sharper, conditional; even sector). [T1] Define the remaining-zero form

$$Q_{\text{rest}}(f) = Q_0(f) - 4F(\gamma)^2 = Q_\zeta(f) - 2F(\gamma)^2,$$

written $\tilde{Q}$ below. Suppose every nontrivial zero other than the one being moved lies on the critical line; where the explicit formula applies, that is a sufficient input for $\tilde{Q} \geq 0$. It is not a necessary one, and $\tilde{Q} \geq 0$ does not follow from positivity of Q_ζ , from positivity of Q_0 , or from a zero-free strip: each of those leaves room for $2F(\gamma)^2$ to exceed the margin. Completing the square, $4F^2 - 4\delta^2 FF'' = (2F - \delta^2 F'')^2 - \delta^4 F''^2$, moves the FF'' allowance into $O(\delta^4)$ and gives

$$\lambda_{\min}(Q_\delta) \geq -4K(L, \gamma)\delta^2 - \left(\frac{2}{5} + \frac{16}{3}e^{2L\delta}\right)L^5\delta^4,$$

where $K(L, \gamma)$ is the exact value of the extremal problem,

$$K(L, \gamma) = \sup_{\substack{f \text{ real, even, } \text{supp} f \subset [-L, L] \\ \|f\|_2=1}} |F'(\gamma)|^2 = \int_{-L}^L u^2 \sin^2(\gamma u) du, \quad (5)$$

attained by $f \propto u \sin(\gamma u)$, with closed form

$$K(L, \gamma) = \frac{L^3}{3} - \left[\frac{L^2 \sin 2\gamma L}{2\gamma} + \frac{L \cos 2\gamma L}{2\gamma^2} - \frac{\sin 2\gamma L}{4\gamma^3} \right].$$

Corollary 9 (γ -uniform form). [T1] Since $\sin^2 \leq 1$ gives $K(L, \gamma) \leq \int_{-L}^L u^2 du = 2L^3/3$ for every γ ,

$$\lambda_{\min}(Q_\delta) \geq -\frac{8L^3}{3}\delta^2 - O(\delta^4),$$

which is the form to quote when γ is not fixed. As $\gamma \rightarrow \infty$, $K(L, \gamma) \rightarrow L^3/3$, so the uniform constant overshoots the large-ordinate value by exactly the factor 2 coming from the average of $\sin^2$.

Remark 10. $4K(L, \gamma)$ is sharp for the derivative-evaluation step: by (5) the supremum is attained, so no smaller δ^2 coefficient is available in that step. Three claims must be kept apart. **(a)** $K(L, \gamma)$ is the exact squared norm of the derivative-evaluation functional $f \mapsto F'(\gamma)$ on the normalised admissible sector. **(b)** $f \propto u \sin(\gamma u)$ attains that norm in the stated space. **(c)** Neither implies that the minimiser of the *entire* Weil form attains the same sensitivity coefficient — attaining the derivative norm need not minimise Q_δ . So $4K$ is not claimed to be a proved sharp coefficient for the optimised minimum eigenvalue; the near-saturation measured in §4.3 is T2. The constants $8L^3/3$ and $4L^3/3$ appearing below are respectively the γ -uniform upper bound and the $\gamma \rightarrow \infty$ limit of the same quantity, and ratios reported against them should be read as such.

4.3 How close the measured constant comes

[T2, src/task_final_checks.py, data/stage3_exact_constant_K.csv]

The planted ordinate is $\gamma_1 = 14.1347$. Measured $C(L)$ against the exact constant $4K(L, \gamma_1)$ of Theorem 8, and for reference against the two constants of Corollary 9:

L	0.8	1.0	1.3	1.6	2.0
measured $C(L)$	0.3108	0.8530	2.4987	4.9218	10.083
$4K(L, \gamma_1)$	0.7419	1.3427	3.1158	5.1130	10.652
$C/4K(L, \gamma_1)$	0.419	0.635	0.802	0.963	0.947
$C/(8L^3/3)$ (<i>uniform</i>)	0.228	0.320	0.427	0.451	0.473
$C/(4L^3/3)$ ($\gamma \rightarrow \infty$)	0.455	0.640	0.853	0.901	0.945

Against the exact constant the re-optimised minimiser reaches 96.3% at $L = 1.6$.

How much the $\sin^2$ weight matters, measured. The deviation $|K(L, \gamma_1)/(L^3/3) - 1|$ is

$$17.3\%, \quad 8.68\%, \quad 0.699\%, \quad 6.37\%, \quad 6.38\%, \quad 0.136\%$$

at $L = 0.6, 0.8, 1.0, 1.3, 1.6, 2.0$. It is *not* monotone: the bracketed terms in (5) oscillate in L with period π/γ , so the correction can be large at one window and negligible at a slightly larger one. At the window where positivity is certified, $L = 0.8$, it is 8.68%; at $L = 2$ it happens to be 0.136%. This is why the γ -uniform constant should not be used as a stand-in for K at a fixed ordinate.

4.4 One law, two settings: the dictionary

[T3, checked numerically in `src/task_final_checks.py`]

The two laws are the same statement under an explicit change of variables. For a curve over $\mathbb{F}_q$ the explicit-formula variable is log of a prime power, so the lattice index n and the ζ variable u correspond by $u = n \log q$. This fixes everything else:

	function field	ζ
window variable	$n = 0, \dots, R$	$u \in [-L, L]$
correspondence	$u = n \log q$, hence window length $R \log q = 2L$	
off-line depth	$a = \log \rho$	$\delta = \operatorname{Re} \rho - \frac{1}{2}$, with $a = \delta \log q$
Rayleigh quotient	sum over n	integral $du = \log q \, dn$: differ by $\log q$

Substituting $R = 2L/\log q$ and $a = \delta \log q$ into Theorem 6 and dividing by $\log q$:

$$-\binom{R+2}{3} a^2 \approx -\frac{R^3}{6} a^2 = -\frac{1}{6} \left(\frac{2L}{\log q} \right)^3 (\delta \log q)^2 = -\frac{4L^3}{3} \frac{\delta^2}{\log q},$$

i.e. exactly the ζ law with the large-ordinate constant $4L^3/3$, divided by the $\log q$ of the last table row. Equivalently both read (window length)³/6 in their own units.

Which factors of 2 cancel. Two enter, in opposite directions, and they cancel:

1. *Pair versus quartet.* Theorem 6 moves a reciprocal *pair*; the ζ counterfactual moves a *quartet*, twice as many zeros — a factor 2 up.
2. *$\sin^2$ averaging.* The ζ constant uses $K(L, \gamma) \rightarrow L^3/3$, half of the $\int u^2 = 2L^3/3$ that the function-field centred moment is the discrete analogue of — a factor 2 down.

Checked on one curve. For $y^2 = x^3 + x$ over $\mathbb{F}_q$, $q = 5$, with $a = 10^{-6}$, the ratio of the exact function-field $\lambda_{\min}$ to the ζ -law prediction $-\frac{4}{3}L^3\delta^2/\log q$ is

$$1.406, \quad 1.195, \quad 1.096, \quad 1.047, \quad 1.024 \quad \text{at } R = 8, 16, 32, 64, 128,$$

converging to 1 like $1 + 3/R$ — which is precisely $\binom{R+2}{3}/\frac{R^3}{6} = (1 + \frac{1}{R})(1 + \frac{2}{R})$, the only discrepancy being the $O(R^2)$ term dropped in passing from the binomial to $R^3/6$. The dictionary is therefore exact at leading order, with a fully accounted correction.

5 The ζ side as a control

[T2; `src/stage3_converge.py`, `src/stage3_connes.py`]

We assemble the window form from Zhu's geometric side in three closed-form bases (two even, one odd), moving everything to u -space by Parseval so no oscillatory quadrature is needed, and summing the archimedean m -series analytically via digamma, Lerch Φ and Hurwitz ζ . The odd-sector pole term $-2P_j P_k$ — the sign flips because $F(-i/2) = -F(i/2)$ for odd f — is derived here.

Gate A. Under RH the explicit formula gives $Q(f) = \sum_{\rho} |F(\gamma_{\rho})|^2$. The assembled geometric side matches the zeros side to 1.63×10^{-7} (even, vanishing at $\pm L$), 2.26×10^{-7} (odd) and 4.23×10^{-3} (full cosine, which jumps at $\pm L$) at $K = 800$ zeros, each decaying at the rate that basis’s smoothness predicts (γ_K^{-3} versus γ_K^{-1}). *Control:* flipping the pole sign inflates the discrepancy to 5.35 and 0.212, so the test discriminates.

Gate B. $\lambda_{\min}(N)$ decreases monotonically in two independent complete even bases to 2.2702×10^{-17} and 2.3357×10^{-17} at $N = 24$, landing on Zhu’s certified $[8.9 \times 10^{-18}, 2.27 \times 10^{-17}]$. Zhu normalises by $Q(f)/\|f\|^2$, the same normalisation used here. We report the inequality, not the matching digits: the sequence is still falling.

Connes’s window. At $2L = \log 13$, $N = 36$, 140 digits: $\lambda_{\min} = 8.977 \times 10^{-52}$, simple and even, with the ground state’s Fourier zeros matching γ_1 to 7.2×10^{-48} and tracking roughly $N/2$ leading zeros. This reproduces at smaller scale what Groskin [6] computes at $N = 100$ –250.

Simplicity and evenness. [T2, $N = 12$, 40 digits] At $L \in \{0.5, 0.8, 1.0, \frac{1}{2} \log 13\}$ the ground state is simple with relative gap 2×10^4 to 4×10^5 and *even* in every case, the odd sector sitting 2–5 orders above. The $L = 0.8$ entry is consistent with Zhu’s certificate; the other three carry no external certificate and are T2 only.

6 Detection onset

[T2; src/stage3e_certify.py, data/stage3e_certified_onset.csv]

Precedent. Bombieri [1, §13] already ran this experiment in 2000: the first $N \leq 160$ zeros of ζ together with “a fictitious zero ρ_0 off the critical line, together with their images by complex conjugation and reflection” — the same quartet — at $\rho_0 = 0.52 + 3.14i$, on windows $E = [-t, t]$, and found a critical value t_c beyond which the negative eigenvalue stays bounded away from 0. He also separates even and odd eigenfunctions. Our contribution is to resolve that critical window in δ and in basis size, and to identify it with Zhu’s bandwidth $T^*(L) = 2\pi e^{2L}$.

The certification asymmetry. A negative Rayleigh quotient from *any* trial function certifies indefiniteness. A finite basis that finds none certifies nothing. We therefore grow the basis (low block $\cup$ resonant block $\{k : w_k \approx \gamma\}$) and report, for each planted ordinate, the smallest window at which *some* trial function goes negative. The geometric side supplies the on-line part exactly; truncating the zeros sum omits positive mass $\sim 10^{-6}$ against a signal $\sim 10^{-27}$ and cannot certify anything near the onset.

planted at	γ_n	$L_{\text{pred}} = \frac{1}{2} \log \frac{\gamma}{2\pi}$	onset, small basis	onset, largest basis
γ_1	14.13	0.405	1.3×	1.3×
γ_5	32.94	0.828	1.0×	1.0×
γ_{10}	49.77	1.035	not found	1.0×
γ_{30}	101.3	1.390	not found	0.85×
γ_{80}	201.3	1.733	not found	1.15×

The onset tracks L_{pred} within about $\pm 30\%$ and moves *earlier* as the basis grows. Rows whose magnitude falls below the measured noise floor (the geometric control is PSD by construction, so a negative value there *is* the floor) are excluded and flagged in the data file.

What detection actually needs — and what it does not. Two conditions are visible in the data: the window must reach the ordinate, $L \gtrsim L_{\text{pred}}(\gamma_n)$, and the basis must *contain modes resonant with γ_n* , i.e. indices near $k^* \approx \gamma_n L / \pi$ — which is what the resonant block supplies. It does *not* require a basis large enough to annihilate the first n zeros. The dimension needed is far below the top mode index: γ_{30} is detected with dimension 46 and top index 45, and γ_{80} with dimension 59 and top index ≈ 134 , whereas annihilating the first n zeros in a contiguous basis would need $N \approx 2n$, i.e. 60 and 160. The $N/2$ rule of §5 governs a different question — how many leading zeros a *contiguous* basis can annihilate — and is not the detection criterion. Detection only needs some f in the subspace with $Q_{\text{geom}}(f) + 2F(\gamma)^2 < 4\delta^2(F'^2 + FF'')$, which the sparse “low block $\cup$ resonant block” subspace achieves at a small fraction of the dimension a contiguous basis would need.

6.1 A certified four-mode witness for an added quartet

[T1 (exact rational interval arithmetic); extensions/planted-witness-consolidation-2026-10-08/]

Everything above in this section is T2: a floating-point search. One point of it is now certified. Fix $L = \frac{4}{5}$, $\gamma = 14$ exactly, and the four quarter-wave modes $\phi_k(x) = \cos((2k+1)\pi x/2L)/\sqrt{L}$, $k = 0, \dots, 3$, extended by 0; these vanish at $\pm L$ because $w_k L = (2k+1)\pi/2$, and are orthonormal. Let

$$f = \sum_{k=0}^3 x_k \phi_k, \quad x = \frac{(99815221908, -354082045264, 928280739889, 54384691304)}{10^{12}},$$

a frozen rational vector with $\|f\|_2^2 = 99999999999981276942897/10^{24}$ (not renormalised; the form is homogeneous of degree 2, so the sign is scale-invariant). For $\delta > 0$ write $Q_{\text{add},\delta}(f) = Q_\zeta(f) + 4(A_\delta^2 - B_\delta^2)$ with $A_\delta = \int f(u) \cos(\gamma u) \cosh(\delta u) du$ and $B_\delta = \int f(u) \sin(\gamma u) \sinh(\delta u) du$; the same quadratic form after *adding* Bombieri’s quartet at $\frac{1}{2} \pm \delta \pm i\gamma$ to the zero multiset, the true zeros left in place.

Theorem 11 (certified four-mode witness). [T1] *With the data above,*

$$Q_\zeta(f) \in [0.0051242393314170015, 0.0052229400457328018] > 0,$$

$$Q_{\text{add},2/5}(f) \leq -0.014510765749822711 < 0,$$

and $Q_{\text{add},\delta}(f) \leq -0.0024549095780663143 < 0$ for every $\delta \in [\frac{1}{4}, \frac{49}{100}]$.

All three are enclosures in exact rational interval arithmetic with outward rounding on a 2^{-260} dyadic grid; π , $\log$, $\exp$, $\sin$, $\cos$ and $\sqrt{}$ are truncated series with proved remainders, and no libm value enters the certified path. The third claim is a union of 256 *abutting* closed cells of width $3/3200$, each enclosure valid for every δ in its cell — not a grid of samples. A single interval evaluation over the whole δ -box returns $+0.0279$ and certifies nothing, because δ occurs independently in δ , δ^2 , $\cosh(\delta L)$ and $\sinh(\delta L)$; that failed attempt is kept as a dependency-inflation control.

How the geometric side was enclosed. This is what Limitation 4 said could not be done, so the route matters. Writing $C(u) = \int f(x)f(x-u)dx$ and using $\text{Re} \psi(\frac{1}{4} + \frac{ia}{2}) + \gamma_E = \sum_{m \geq 0} [\frac{1}{m+1} - \frac{2s_m}{s_m^2 + a^2}]$ with $s_m = 2m + \frac{1}{2}$, the archimedean term becomes a series of *elementary* integrals,

$$\text{Arch}(f) = -\gamma_E C(0) + \sum_{m \geq 0} \left[\frac{C(0)}{m+1} - 2 \int_0^{2L} e^{-s_m u} C(u) du \right],$$

with no digamma and no Lerch transcendent. Three facts make this exact. The prime sum is *finite and complete*: C is supported in $|u| < 2L$, and $e^{8/5} < 5$ is certified, so $n \in \{2, 3, 4\}$

exhausts it. The boundary term at $u = 2L$ vanishes *identically*, since the two translates of $\text{supp } f$ meet in the single point $\{L\}$ — an identity, not the numerical $|C(2L)| < 9 \times 10^{-78}$ that merely confirms it. And γ_E is eliminated, not enclosed, by $H_M - \gamma_E = \log(M+1) - T$ with $T \in [\frac{1}{2(M+1)} - \frac{1}{6M^2}, \frac{1}{2M}]$ and $M+1 = 2^{17}$, so $\log(M+1) = 17 \log 2$ exactly. The truncation at $M = 2^{17} - 1$ costs $[\frac{3}{4}C(0) + \frac{1}{2} \sup |C'|]/(M - \frac{3}{4}) = 4.94 \times 10^{-5}$, where $\sup_{[0,2L]} |C'| \leq \sum_k (|A_k|w_k + |B_k| + |B_k| \cdot 2Lw_k) \leq 11.4368$ by the triangle inequality on the closed form — a proof at every point, not a sampled maximum. The margins exceed that allowance by $294 \times$ (point) and $50 \times$ (interval-wide).

Independently reimplemented. The point inequality was then re-derived and re-evaluated by a second program sharing no code: it computes the *same* archimedean series in closed form through ψ and ψ' at the four points $z_k = \frac{1}{4} - \frac{iw_k}{2}$, where γ_E cancels identically and no $1/M$ tail arises, and it obtains $A_\delta - iB_\delta$ by a complex-sinc route rather than the cosh/sinh split. Its enclosure $Q_\zeta(f) \in [0.0051612157, 0.0052247976]$ overlaps the first (neither contains the other, as two different error budgets should do), its $A_{2/5}, B_{2/5}$ agree to 77 digits, and it returns $Q_{\text{add},2/5}(f) \leq -0.014508908225396348 < 0$ end to end. This removes the single-implementation risk; it is not external review.

What it does and does not say. It certifies that a four-mode window detects an *added* off-line quartet at displacement $\delta \in [\frac{1}{4}, \frac{1}{2})$ and height $\gamma = 14$, and that the same window is positive for the actual zero set. It says nothing about whether such a quartet exists, gives no restriction on the zeros of ζ , is no new RH criterion, and is not an optimal threshold: $\delta \in (0, \frac{1}{4})$ is uncovered. Nor does it extend in height — $\|Per_\gamma\|^2$ falls from 3.60×10^{-2} at $\gamma = 14$ to 1.13×10^{-7} at $\gamma = 280$, so $K_{\text{eff}} \rightarrow 0$ and no uniform-in- γ statement follows. The count-preserving variant $Q_{\text{rep},\delta} = Q_\zeta - 2F(\gamma)^2 + 4(A_\delta^2 - B_\delta^2)$ is a different object; here the two differ by $2F(14)^2 = 3.41 \times 10^{-25}$, far below every margin, and $F(14) = 4.13 \times 10^{-13}$ is carried throughout, never set to zero.

Weak δ -dependence, explained. Detection is $C(L)\delta^2 > \lambda^*(L)$. By Zhu’s Landau–Widom law $\lambda^*(L)$ falls super-exponentially in L while $C(L) \sim L^3$ grows polynomially, so a factor in δ^2 shifts the crossing only logarithmically. Measured: a tenfold change in δ moves the onset from $L = 0.5$ to $L = 0.6$.

7 The recorder split

[T2; `src/task2_inertia_reconcile.py`]

Write $T_R = A - 2M$. In the function field the pole part of A is $q^{|n|/2} + q^{-|n|/2} = u_i w_j + w_i u_j$, an exact rank-2 *hyperbolic* plane whose scale grows like $q^{R/2}$ against a fixed $2g \cdot I$; so A is eventually indefinite under both conventions tested, M is indefinite, and $S = T_R$ is PSD. This is structural.

On ζ windows it is not. With $A = \text{Pole} + \text{Arch} - \log \pi \cdot I$, $2M = \text{Prime}$ one finds A with negative index 1 for $L \leq 0.8$ and 2 at $L = \frac{1}{2} \log 13$, identical in all three bases. But grouping $-\log \pi$ with the primes instead makes **both A and M positive definite** at $L \leq 0.8$. The reason is structural: for even f the ζ pole term is $h(i/2) + h(-i/2) = 2F(i/2)^2 \geq 0$, rank one and *definite*, not hyperbolic. There is no hyperbolic plane on the ζ side to find. We therefore state the convention ($A = \text{Pole} + \text{Arch} - \log \pi \cdot I$) and claim Block F only for the function field.

8 Relation to prior work

Hallouin–Perret [8] is the function-field foundation: Proposition 5 is the window Toeplitz matrix, Theorem 6 the first-singular-size and kernel generator, Theorem 36(ii) step (i), Lemmas 33–34 the rank rule and parity split used here. **Connes–van Suijlekom** [5] supply the C^* -algebraic proof of step (i) and the Hurwitz step, cite [8] for the function-field connection, and state the simplicity difficulty themselves; their Remark 2.3 is our degeneracy observation in words. **Bombieri** [1] is the precedent for planted ζ experiments (§13) and proves the negative-eigenvalue count (Theorem 8); **Yoshida** [10] first studied the Weil hermitian form on functions supported in $[-t, t]$. **Zhu** [11] supplies the certified $\lambda^*(0.8)$, the normalisation and the bandwidth T^* . **Groskin** [6, 7] supplies the large-scale Galerkin computations and the archimedean tail order. **Connes–Consani–Moscovici** [3], **Connes** [4], **Connes–Consani** [2] and **Suzuki** [9] are the surrounding program. A fuller account, with verbatim quotations and the verification method, is in `PRIOR_ART.md`.

9 Limitations

1. No rigorous lower bound on $\lambda^*(L)$ of our own. Theorem 7 is a *relative* estimate; it becomes an unconditional lower bound only where Zhu’s certificate reaches ($L \leq 0.8$). Replacing $\lambda^*(L)$ by 0 beyond that range would assume the window positivity at issue, and is not asserted (Retraction 18).
2. Theorem 8 is conditional on all other zeros being on the line. Nonnegativity of the remaining-zero form $\tilde{Q} = Q_\zeta - 2F(\gamma)^2$ is not supplied by positivity of Q_ζ , by positivity of Q_0 , or by a zero-free strip.
3. A zero-location restriction alone does not discharge the hypothesis of Theorem 8. For the audited source version and the transfer attempted in `extensions/openai-quasi-rh-audit-2026-10-07/`, no usable estimate was obtained that replaces the remaining-zero positivity hypothesis: the restriction bounds how far a zero may sit from the critical line, not the aggregate window-weighted contribution of all such zeros, and the per-zero bound used there is uniform in the ordinate and therefore not summable as it stands. This is a limitation of that particular estimate, not a general obstruction.
4. Interval certification of the *general* pipeline covers the zeros-side assembly only: `mpmath`’s interval context has no digamma and no Lerch transcendent, so the geometric side cannot be interval-evaluated there, and what is certified instead is the $\text{LDL}^\top$ witness identity $v^\top Z v = D_j$ and a measured noise floor. *Amended*: at the single point of §6.1 this limitation no longer applies. Re-expressing the archimedean term as a series of elementary integrals removes both special functions, so the geometric side *is* interval-evaluated there, in exact rational arithmetic and without `mpmath`. The route is specific to that window and basis and does not lift to the sweeps.
5. Onset non-detections are limited by the subspaces actually searched: the largest had dimension 59 with top mode index ≈ 134 . A non-detection bounds that search, not the method — and not the basis either: in `src/zeta_window.py` the frequency $(2k + 1)\pi/2L$ depends on k and L only, never on N , so at fixed L the quarter-wave bases are *nested* and zero-padding the witness of §6.1 to dimension 8 or 16 gives the identical function and the identical certificate (Retraction 20).
6. $C(L)$ is measured at finite δ and finite basis; the saturation $C/(4L^3/3) \rightarrow 1$ is observed, not proved.
7. The ζ -side numbers depend on the planted ordinate, the basis and the truncation. Theorem 11 is one ordinate ($\gamma = 14$), one window ($L = \frac{4}{5}$) and one basis (dimension 4); it is

not uniform in any of them.

8. The region where the Taylor correction is controlled and the region where §6.1 certifies negativity **do not overlap**. The certified ratio $R/(2K_{\text{eff}}\delta^2)$ is 0.3301 at $\delta = \frac{1}{10}$ and 0.8045 at $\frac{3}{20}$, crosses 1 between $\delta = 0.166$ and 0.167, and reaches 8.534 at $\delta = \frac{2}{5}$; the certificate is negative only for $\delta \geq \frac{1}{4}$. The gap $\delta \in (0.166, 0.25)$ is covered by neither, and the $\theta < \frac{1}{4}$ route does not reach into it (Retraction 19).
9. No formal verification. The arithmetic of §6.1 is exact-rational and interval-sound and its analytic steps are proved in `extensions/planted-witness-consolidation-2026-10-08/`, but CPython, its `int/Fraction` implementation and the operating system remain computational dependencies; there is no Lean or Coq artifact. The second implementation removes the single-implementation risk, not the single-author risk: it is not external peer review.

10 Traps this project documents

The retraction log (Appendix A) is part of the result. The reusable lessons: **(a)** truncating the zeros sum biases the form *downward* and manufactures false detections; **(b)** moving a zero and adding a zero are two different counterfactuals, both legitimate — Bombieri studies the second — and the sharp law needs the first; **(c)** `float64` returns spurious negative eigenvalues of size 10^{-16} where the true value is exactly 0; **(d)** interval libraries have gaps, and `mpmath`’s `iv.sincpi` is itself broken (it calls a non-existent `ctx.sincpi`); **(e)** a non-detection certifies nothing; **(f)** keeping known-good regression values was the only thing that caught three sector-rename bugs.

A Retractions

Twenty corrections were logged during this work; the full list with evidence is in `REPORT.md`. Those a reader must see:

#	Retraction
1	Frobenius radius computed on P rather than its reverse (0.8 instead of $\leq 1.3 \times 10^{-51}$).
3	A tolerance-based sign test misclassified exact zeros; replaced by an error-based test plus exact rational certificates.
4	H2 as first stated was refuted by my own adversarial pass ; restated with a distinctness hypothesis. The restatement is Theorem 2.
6	“Step (iv) is free” retracted — it had been scored on zero angles, which is the wrong object. At the function level it carries content (Proposition 5).
8–9	A 40-digit run discarded; three sector-rename bugs caught only by regression values.
11	“Low-lying zeros only” retracted as structural . The height sweep had used a truncated on-line side, and non-detections were basis-limited. Redone, $\gamma_{30} \approx 101$ is detected at $0.85 L_{\text{pred}}$. <i>Amended</i> : the added-quartet counterfactual is legitimate in its own right — it is Bombieri’s — and was not the error.
12	γ_{80} “detections” below their own noise floor discarded; later detected at $1.15 L_{\text{pred}}$ with a positive control.
15	“Block F holds on the ζ side” retracted (§7).

#	Retraction
16	Three claimed contributions are published theorems: step (i) is [8, Thm. 36(ii)], the first-singular-size result is [8, Thm. 6], the window matrix is [8, Prop. 5]; the degeneracy observation is [5, Rem. 2.3]; the negative-index count is [1, Thm. 8]; the planted experiment is [1, §13]. An earlier abstract-level novelty check had missed all of these.
17	The $N \gtrsim 2n$ detection rule retracted (§6). It contradicted both the stated limitation and the γ_{80} table row. The γ_{80} detection used dimension 59, top index ≈ 134 , against $2n = 160$; γ_{30} used dimension 46, top index 45, against $2n = 60$. The rule had been imported from §5, where $N/2$ governs a stronger question — how many leading zeros a <i>contiguous</i> basis annihilates. The table row and the limitation were correct; only the rule was wrong.
18	The unsupported replacement of the unperturbed baseline by zero outside the certified range has been withdrawn. Theorem 7 previously added that for $L > 0.8$ it “holds with $\lambda^*(L)$ replaced by 0, using no positivity input at all”. Substituting 0 for $\lambda^*(L)$ in a lower bound requires $\lambda^*(L) \geq 0$, which is the finite-window Weil positivity at issue; at $\delta = 0$ the substituted statement asserts $\lambda_{\min}(Q_0) \geq 0$, i.e. the very input it disclaimed. The preceding relative perturbation estimate is retained, and is now stated as such. The same step was the last link of the chain in §B.4 of <code>notes/proofs.md</code> and is corrected there. No numerical value, no other theorem and no table is affected.
19	“$R \leq 2K_{\text{eff}}\delta^2$ fails at every positive δ” retracted — it was false. The bound <i>holds</i> for small δ : the certified ratio $R/(2K_{\text{eff}}\delta^2)$ is 0.3301 at $\delta = \frac{1}{10}$, 0.8045 at $\frac{3}{20}$, 1.0108 at 0.166, 1.0247 at 0.167, 1.5493 at $\frac{1}{5}$ and 8.5343 at $\frac{2}{5}$. The obstruction is not that the bound never holds but that its region of validity, $\delta \lesssim 0.166$, does not meet the region $\delta \geq \frac{1}{4}$ where §6.1 certifies negativity. The gap is now stated as item 8 of §9 rather than as a failure of the bound.
20	“Dimensions 8 and 16 do not detect” retracted as a statement about the basis. The quarter-wave frequencies depend on k and L only, not on N , so at fixed L the bases are nested and the dimension-4 witness of §6.1 sits inside both larger ones with the identical certificate. Those bases <i>do</i> detect it; only the <i>search</i> failed to find it. The distinction is the same one as Retraction 11.

B Reproduction

Code and data: <https://github.com/innerlightr-wq/weil-window-control>. Python 3.12, `mpmath` 1.3.0, `numpy` 1.26.4, `matplotlib` 3.8+; no `gmpy2` (installing it speeds the high-precision runs substantially). The four scripts of §6.1 are an exception: they use the **standard library only** — no `mpmath`, no `numpy` — and were run under CPython 3.13 on Linux `x86_64`, taking 5m25s (cold certificate, cache deleted), 3m17s (second implementation), ≈ 4 s and ≈ 45 s. All four exit 0; the first prints four gates, all PASS. Full commands, settings and tail parameters are in that directory’s `REPRODUCE.md` and `VERIFICATION_REPORT.md`, the analytic steps in its `ANALYTIC_AUDIT.md`.

Result	Script
§3 genuine	src/stage1.py
§3 planted	src/stage2.py
§3 kernel	src/function_level.py
Thm 6, 7, 8	src/proofs_check.py
C vs Cauchy–Schwarz	src/task34_extra.py
$K(L, \gamma)$, dictionary	src/task_final_checks.py
Gates A, B	src/stage3_converge.py
Connes window	src/stage3_connes.py
Detection onset	src/stage3e_certify.py
Recorder split	src/task2_inertia_reconcile.py
Thm 11	planted-witness-consolidation-2026-10-08/verify_certificate_consolidated.py
second implementation	planted-witness-consolidation-2026-10-08/independent_point_check.py
primitive/cache audits	planted-witness-consolidation-2026-10-08/audit_primitives.py, audit_cache_invalidation.py

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